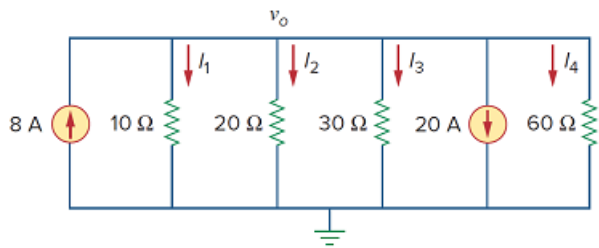


Problem

Find the currents I_1 through I_4 and the voltage v_o in the circuit of Fig. 3.52.

Figure 3.52:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.52 in the textbook.

Write the Kirchhoff's current law equation at node v_o .

$$-8 + \frac{v_o}{10} + \frac{v_o}{20} + \frac{v_o}{30} + 20 + \frac{v_o}{60} = 0$$

$$\left(\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{60} \right) v_o = -12$$

$$\left(\frac{6+3+2+1}{60} \right) v_o = -12$$

$$\left(\frac{12}{60} \right) v_o = -12$$

$$\left(\frac{1}{5} \right) v_o = -12$$

$$v_o = -60 \text{ V}$$

Therefore, the voltage, v_o is $\boxed{-60 \text{ V}}$.

Step 2 of 3

From the Figure 3.52 in the textbook, the current I_1 is,

$$\begin{aligned} I_1 &= \frac{v_o}{10} \\ &= \frac{-60}{10} \\ &= -6 \text{ A} \end{aligned}$$

Therefore, the current, I_1 is $\boxed{-6 \text{ A}}$.

Calculate the current, I_2 .

$$\begin{aligned} I_2 &= \frac{v_o}{20} \\ &= \frac{-60}{20} \\ &= -3 \text{ A} \end{aligned}$$

Therefore, the current, I_2 is $\boxed{-3 \text{ A}}$.

Calculate the current, I_3 .

$$\begin{aligned} I_3 &= \frac{v_o}{30} \\ &= \frac{-60}{30} \\ &= -2 \text{ A} \end{aligned}$$

Therefore, the current, I_3 is $\boxed{-2 \text{ A}}$.

Calculate the current, I_4 .

$$\begin{aligned} I_4 &= \frac{v_o}{60} \\ &= \frac{-60}{60} \\ &= -1 \text{ A} \end{aligned}$$

Therefore, the current, I_4 is $\boxed{-1 \text{ A}}$.